💰 Personal Expense Tracker (MERN Stack)

Project Overview

This is a full-stack personal finance application built on the **MERN** (MongoDB, Express, React, Node.js) stack. It allows users to securely track, visualize, and manage their income and expenses in a clean, responsive dashboard.

Key Features

- **Secure Authentication**: User registration and login using JWT (JSON Web Tokens) and secure password hashing (`bcryptjs`).
- **Transaction Management**: Add, view, and delete income and expense records.
- **Data Visualization**: Dashboard displays financial summaries using Pie, Bar, and Line charts (powered by Recharts).
- **Data Export**: Download all income and expense details as a `.xlsx` Excel file.
- **Profile Management**: Upload and manage a profile picture (using Multer for file handling).

Technology Stack

Component	Technology	Role
Frontend	React (Vite), Tailwind CSS	UI components, client-side routing, global state management (Context API).
Backend	Node.js, Express.js	REST API, Business Logic, Middleware (Auth, Upload).
Database	MongoDB (Mongoose)	Data persistence, schema definition, and querying.
Security	JWT, `bcryptjs`	Authentication, Authorization, and secure password storage.

🚀 Setup and Installation

1. Prerequisites

You must have **Node.js** (version 18+) and **npm** installed on your system.

2. Backend Setup (API Server)

The backend handles the API and database connection. It runs on `http://localhost:8000`.

1. Navigate to the `backend` folder in your terminal.

```
cd backend
```

2. Install the required Node.js dependencies:

```
npm install
```

3. Database Configuration (MongoDB Atlas):

You need to set up a free MongoDB Atlas cluster to get your connection string.

a. Go to https://www.mongodb.com/. Log in or create an account.

b. Create a New Project and set up a Shared Cluster (Free Tier).

c. Set up a database user (remember the username and password).

d. Configure Network Access to allow connections (e.g., allow access from anywhere for development).

e. Go to the Database section, click Connect, select Drivers, and copy the connection string.

4. Configure Environment Variables (.env file):

Create a file named .env in the root of your backend directory and paste the following, replacing the bracketed placeholders:

```
PORT=8000
CLIENT_URL="http://localhost:5173"
```

```
MONGO_URI="[YOUR_MONGO_CONNECTION_STRING]"
JWT_SECRET="[YOUR_SECRET_KEY]"
# CLOUDINARY_CLOUD_NAME=[Your Cloud Name]
# CLOUDINARY_API_KEY=[Your API Key]
# CLOUDINARY_API_SECRET=[Your API Secret]
```

5. Finalizing the Connection String:

In the MONGO_URI you pasted, replace the <password> placeholder with the actual database user's password you created in step 3c.

6. Generate JWT Secret:

For a strong JWT_SECRET, run the following command in your terminal and paste the output into the .env file:

```
node -e
"console.log(require('crypto').randomBytes(64).toString('hex'))"
```

7. Start the Backend Server:

Run the following command to start the server in development mode (it will automatically restart on code changes):

```
npm run dev
```

3. Frontend Setup (React Application)

The frontend application uses the Vite development server and runs on `http://localhost:5173`.

1. Navigate to the main `frontend` folder (or the project root where your frontend `package.json` is located).

```
cd ../frontend # (if you are still in the backend folder)
```

2. Install the required React dependencies:

```
npm install
```

3. Start the Frontend Server:

Run the following command to start the React development server:

```
npm run dev
```

The application should automatically open in your default web browser at `http://localhost:5173`. You can now register a new user and start tracking your finances!